Keysight PROPSIM

Quick Start Guide for the PROPSIM FS16 radio channel emulator F8820B

Quick Start Guide

KEYSIGHT TECHNOLOGIES

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This is a general hazard sign, which is used to draw the user's attention to warnings and cautions

WARNING! The WARNING! sign denotes a hazard. It identifies conditions or practices that could result in personal injury or death.

CAUTION The CAUTION sign denotes a hazard. It identifies conditions or practices that could result in damage to the equipment or to other property.

The following safety notices can be found on Keysight equipment:

WARNING!

Before this instrument is switched on, make sure it has been properly grounded through the protective conductors of the ac power cable to socket outlets provided with protective earth contacts.

Any interruption of the protective (grounding) conductor, inside or outside the instrument, or disconnection of the protective earth terminal can result in PERSONAL INJURY.

Install the device to a location allowing easy access to the device main power switches. Ensure the installation site has sufficient clearances around the device cabinet.

If this instrument is used in a manner not specified by Keysight Technologies, the protection provided by the instrument may be impaired.

There are many points in the instrument that can, if contacted, cause personal injury. Be extremely careful. Any adjustments or service procedures that require operation of the instrument with the protective covers removed may be performed only by trained service personnel.

Do not cover the device. Make sure that the emulator has unrestricted airflow for the fan and ventilation openings.

The emulator should be carried by at least two people due to its size and weight.

The main switch of the device is equipped with over current protection. The switch bounces back to OFF position in case of over current. If this happens and the switch refuse to stay in ON position, do not try to force the switch.

CAUTION

Before this instrument is switched on, make sure its primary power circuitry has been adapted to the voltage of the ac power source.

Failure to set the AC power input to the correct voltage could cause damage to the instrument when the AC power is plugged in. See chapter 2.2 for more information.

Safety Earth Ground

The Mains wiring, and connectors shall be compatible with the connector used in the premise electrical system. Failure to ensure adequate earth grounding by not using the correct components may cause product damage and serious injury.

Before Applying Power

If this product is not used as specified, the protection provided by the equipment could be impaired. This product must be used in a normal condition (in which all means for protection are intact) only.

If an instrument's shipping case handle is damaged, you should replace it immediately. Damaged handles can break while you are moving or lifting the instrument and cause personal injury or damage to the instrument.

This instrument has auto-ranging line voltage input. Be sure the supply voltage is within the specified range. This product is designed for use in Installation Category II and Pollution Degree 2 per IEC standards.

Ventilation Requirements: When installing the product in a cabinet, the convection into and out of the product must not be restricted. When installing product to a cabinet with restricted air flow, forced convection must be used.

The measuring terminals on this instrument are designed to be used with external signals described in Measurement Category I, but NOT with external signals described in Categories II, III, and IV. The input of this instrument cannot be connected to the mains.

Servicing

These servicing instructions are for use by qualified personnel only. To avoid electrical shock, do not perform any servicing unless you are qualified to do so.

The opening of covers or removal of parts may expose dangerous voltages. No operator serviceable parts inside. Refer servicing to qualified personnel. To prevent electrical shock, do not remove covers.

Instructions for Use

This product has been designed and tested in accordance with accepted industry standards and has been supplied in a safe condition. The documentation contains information and warnings that must be followed by the user to ensure safe operation and to maintain the product in a safe condition.

This is a Safety Class 1 Product (provided with a protective earthing ground incorporated in the power cord). The mains plug shall only be inserted in a socket outlet provided with a protective earth contact. Any interruption of the protective conductor inside or outside of the product is likely to make the product dangerous. Intentional interruption is prohibited.

Install the instrument so that the detachable power cord is readily identifiable and is easily reached by the operator. Alternatively, an externally installed switch or circuit breaker (which is readily identifiable and is easily reached by the operator) may be used as a disconnecting device.

The power cord is connected to internal capacitors that may remain live for 30 seconds after disconnecting the plug from its power supply. The detachable power cord is the instrument disconnecting device. It disconnects the mains circuits from the mains supplies before other parts of the instrument. The front panel switch is only a standby and ON switch and is not a LINE switch (disconnecting device).

The instruction documentation symbol. The product is marked with this symbol when it is necessary for the user to refer to the instructions in the documentation.

This symbol indicates separate collection for electrical and electronic equipment, mandated under EU law as of August 13, 2005. All electrical and electronic equipment are required to be separated from normal waste for disposal (Reference WEEE Directive, 2002/96/EC.

This symbol indicates that the power line switch is in the STANDBY position.

I	This symbol indicates that the power line switch is ON.
O	This symbol indicates that the power line switch is in the OFF position.
CE	The CE mark is a registered trademark of the European Community. (If accompanied by a year, it is when the design was proven.)
c SP us	The CSA mark is a registered trademark of CSA - International. This instrument complies with Canada: CAN/CSA-C22.2 No. 61010-1-12.
ICES/NMB-001	This ISM device complies with Canadian ICES-001. Cet appareil ISM est conforme a la norme NMB du Canada.
40	China RoHS regulations include requirements related to packaging, and require compliance to China standard GB18455-2001.
KC	South Korean Certification (KC) mark; includes the marking's identifier code which follows this format: MSIP-REM-YYY-ZZZZZZZZZZZZZZ.
	This symbol indicates compliance with China RoHS regulations for paper/fiberboard packaging.

Important information concerning product disposal and recycling

When this crossed-out garbage bin symbol is attached to a product, it may not be mixed with general waste in case of product disposal.

There is a separate collection system for used electronic products in accordance with legislation that requires proper treatment, recovery and recycling.

In case of product disposal, customers of this product (PROPSIM) in the member states of the EU, Switzerland and Norway may return their used products free of charge to a retailer to be disposed properly.

For countries not mentioned above, please contact your local authorities for the correct method of disposal.

By doing so you will ensure that your disposed product undergoes the necessary treatment, recovery and recycling, and thus will prevent potential negative effects on the environment and human health.

For more information, please contact Keysight.

ABBREVIATIONS

Name	Description
ATE	Automatic Test Equipment
BS	Base Station
CIR	Channel Impulse Response
FDD	Frequency-division duplex
GUI	Graphical User Interface
HDD	Hard Disk Drive
MANET	Mobile ad hoc network
MIMO	Multiple Input Multiple Output
MISO	Multiple Input Single Output
MS	Mobile Station
RF	Radio Frequency
RMS	Root Mean Square
RX	Receiver
SCM	Spatial Channel Model
SCME	SCM Extended
SFP	Small form-factor pluggable
SIMO	Single Input Multiple Output
SISO	Single Input Single Output
SNR	Signal to Noise Ratio
TX	Transmitter
UDP	User Datagram Protocol
VAC	Voltage with Alternate Current
VDT	Virtual Drive Test
WLAN	Wireless LAN

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1 INTRODUCTION

This document is the quick start guide of the PROPSIM radio channel emulator. The quick start guide covers the initial setup of the product and the operation and use of the PROPSIM applications.

A multitude of different test scenarios, channel emulations, can be run on the PROPSIM. Thus, explaining each and every emulation setup possibility is outside the scope of this document. This quick start guide focuses on the most common use cases of the PROPSIM and includes explanations and screenshots of the necessary configuration steps. The document can be used to easily find a solution to a basic configuration task, but also as a reference material when designing more complex emulations.

The guide is divided into six main chapters:

- "Introduction" gives background information about the PROPSIM emulator in general and explains the most important features of the product. It also introduces the physical connectors that can be found in the front panel of the product.
- "Getting started" provides information on how to get the PROPSIM up and running quickly and how to shut down the device.
- "Introduction of the GUI" describes the main elements of the PROPSIM Graphical User Interface.
- "Basic functions" explains how to use Scenario Wizard and run typical radio channel emulations.
- "Technical specification" lists the technical data related to the operating environment.
- "Maintenance" provides cleaning instructions and shipment for service information.

More detailed information about PROPSIM can be found in the User reference documentation, included in the PROPSIM installation. To access all the documentation online, please log in to myKeysight (http://www.keysight.com/my).

The quick start guide includes a number of screenshots taken from the PROPSIM Graphical User Interface (GUI) and references to these figures. To help navigation in the GUI and to further clarify the explanations, the important details in the screenshots are highlighted with green color.

When mentioned in this quick start guide, term "PROPSIM" applies to PROPSIM FS16 (F8820B) device unless otherwise stated.

1.1 Basic Terminology, PROPSIM Essentials

Radio channel model is well defined propagation condition (typically multipath fading, Doppler) between one transmitter and receiver. For example, standardized channel models such as Vehicular, Pedestrian, Hilly Terrain, SCME Urban Micro, SCME Urban Macro etc. Please note there is a set of 3GPP standardized radio channel models included in PROPSIM standard toolkit for conductive testing.

Test Scenario is a ready to run PROPSIM emulation file-set that can be used for testing as such, without a need for user to define and create test scenario and emulation into PROPSIM. In advanced test scenario there could be multiple transceivers all mobile simultaneously in dynamic propagation conditions (multipath, Doppler, path loss and interference) which could include set of standardized channel models or channel conditions measured from field with special test UE or test receiver.

Virtual drive testing and indoor testing (VDT) is generally a field-like, end-to-end testing for devices and live networks, but in laboratory conditions. Test scenarios are based on drive test or indoor test data utilization (field to lab with VDT Toolset). Test scenarios can also be fully user defined with other PROPSIM modeling tools (Standard tools, GCM tool, WLAN tool).

Test scripts are test configuration files for PROPSIM and external automation software, providing testing capabilities but do not provide ready results or reports for users. External automation software is not in scope of PROPSIM standalone channel emulator options. Please contact Keysight for availability of complete test systems.

Test Cases are ready to run test sequence on external automation software that provides result pass/fail or performance metrics in ready report. External automation software is not in scope of PROPSIM standalone channel emulator options. Please contact Keysight for availability of complete test systems.

Fading channel emulation consist both of slow and fast fading. It is implemented partially at RF channel and partially at digital channel by PROPSIM channel emulator.

RF channel in the PROPSIM channel emulator consists of an input - output pair, with the connectors located at the front panel of the device. RF channel operates at the user selected radio frequency. Radio channel phenomena between the input and the output are modelled in the **Digital Channel**.

Digital channel, logical channel applies the radio channel model to the signal travelling from RF input to RF output. Digital channel shapes the signal to emulate multipath fading effects as delay, Doppler, phase and amplitude response. Number of logical channels needed for the single test setup typically depends on the antenna count of devices under the test (MIMO dimensions). Number of logical channels available in the PROPSIM device depends on the device configuration and selected fading parameters.

1.2 PROPSIM

The wireless environment imposes many constraints and limitations on the performance of wireless telecommunication systems. Radio channel propagation characteristics, such as attenuation, shadowing, fast fading, variable delays, Doppler effect, noise and interference cause severe degradation to all wireless transmission. Thus, the development of wireless communication systems requires rigorous testing to ensure that the products are able to operate under demanding propagation environments.

The traditional field-testing of wireless systems is generally labor intensive, time-consuming and expensive. Furthermore, even under the same test setups and test scenarios the test results are typically non-repeatable, since the propagation environment uncertainties plus external noise and interference affect the results.

The PROPSIM is a radio channel emulator that enables recreating the wireless channel propagation effects in a controlled laboratory environment. It is a solution for performing a realistic and accurate emulation of all typical radio channel propagation effects such as multipath propagation, fast fading, dynamic delays, attenuation, noise, interference and shadowing. The PROPSIM supports multiple channels, wide bandwidth, high dynamic range and channel emulation with very high accuracy. The physical radio channel characteristics can all be emulated independently on PROPSIM, supplementing or even replacing traditional field-testing. PROPSIM F16 (F8820B) is shown in Figure 1.

Figure 1 PROPSIM FS16 F8820B configurations: BRF1, BRF2, and BRF1 + BRF2 channel units

PROPSIM has been developed to fulfil the requirements of both present and future wireless communication systems. With extremely high RF performance it provides superior performance and feature set for radio interface testing of current cellular communication systems, such as 5G, LTE, LTE-A, WLAN, WCDMA, GSM, TD-SCDMA and EV-DO / CDMA2000. Additionally, the features of the PROPSIM enable testing of the next generation communication systems.

1.3 Channel emulation concept

The PROPSIM is a versatile state of the art channel emulator. It emulates the radio channel excluding transmitter and receiver and is thus independent of system technology or modulation. The PROPSIM supports all major wireless standards and signal types in a broad frequency range. It supports the development of most demanding wireless applications, such as beamforming with large MIMO configurations, software defined radios and MANET (Mobile ad hoc network).

In a typical test scenario, the transmitter and receiver to be tested are connected to the PROPSIM, which then emulates a wireless propagation environment, replacing the real radio channel, as illustrated in Figure 2. The PROPSIM uses real-world signals generated by the external test equipment as an input.

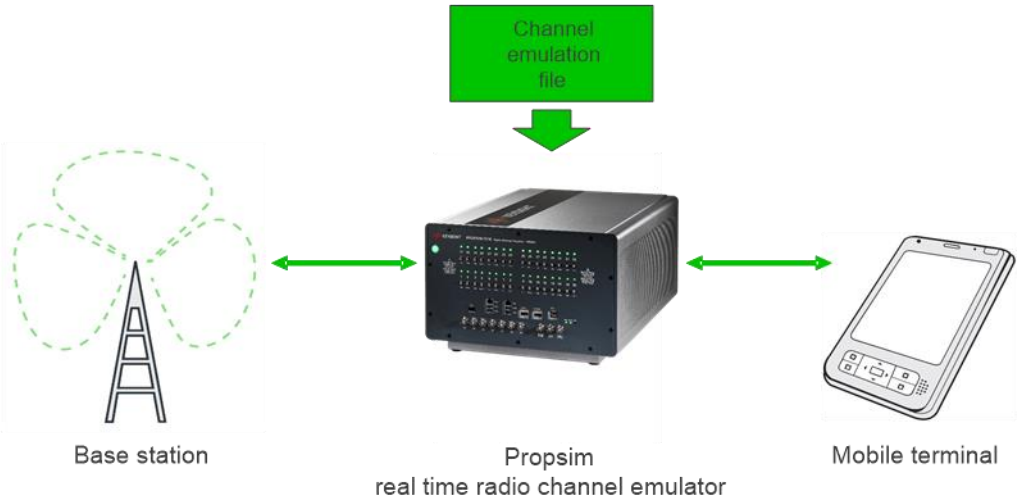


Figure 2 Replacing the radio channel with PROPSIM

The overall flexibility and extensive number of configuration options of the PROPSIM allow it to run a number of different types of tests and emulations. The emulations are based on preconfigured channel emulation files, stored in the PROPSIM. New emulation files can be added any time by the users. The file-based emulation approach ensures full repeatability and controllability of the emulations.

The PROPSIM includes pre-stored standard channel emulations (emulation connection diagram with channel models according to e.g. 3GPP standards). In basic use, the user only needs to load the emulation and adjust the power levels of the PROPSIM. For advanced use, all versatile channel modeling tools for creating user defined channel models and to modify pre-defined standard channel models are available.

The PROPSIM includes an easy to use GUI that guarantees quick emulation setup. The GUI can be used to control all aspects of the emulator functionality. It ensures that all emulation setup and configuration tasks require minimal amount of manual work.

The PROPSIM platform comes with the Standard Toolset software that includes applications for creating channel models and emulations. The channel models are stored in the PROPSIM as pre-calculated files.

1.4 Physical connectors and LEDs

All external connectors and LEDs of the PROPSIM are found in the emulator front panel, as shown in Figure 3 (BRF1) and Figure 4 (BRF2).

Figure 3 PROPSIM FS16 F8820B with BRF1 channel units

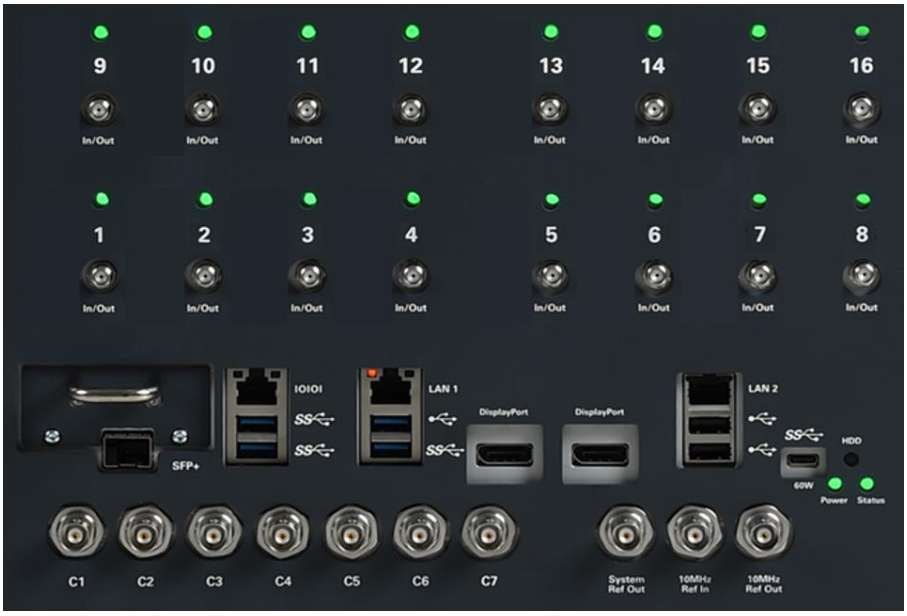


Figure 4 PROPSIM FS16 F8820B with BRF2 channel units

The PROPSIM FS16 can equipped with following channel unit configurations:

- BRF1 configuration: up to 16 Bidirectional TRX ports or 16 Unidirectional TX and RX ports (FR1)
- BRF2 configuration: up to 16 Bidirectional TRX ports (FR3)
- BRF1 + BRF2 configuration: up to 8 Bidirectional TRX ports or 8 Unidirectional TX and RX ports (FR1) and up to 8 Bidirectional TRX ports (FR3)

Table 1 lists the physical connectors and LEDs found in the front panel of the PROPSIM FS16.

Table 1 PROPSIM FS16 connectors and LEDs

Name	Description
RF IN/OUT	RF In / Out connectors, even numbered RF connectors in BRF1 channel units, all RF connectors in BRF2 channel units
RF OUT	RF Out connectors, BRF1 (odd numbered RF connectors)
SFP+	Digital data interface (for future use)
IOIOI	Digital communication interface (for future use)
LAN1, LAN2	Gigabit LAN, 10/100/1000 Base-T connector for LAN connectivity (2)
USB	USB A connectors (6) for external keyboard, mouse and memory. USB C connector (1) for Keysight smart accessories. Note, USB cable length should be < 3m.
DisplayPort	DisplayPort connectors (2) for external displays
C1	Sync in
C2	Sync out
C3	Trigger in (Signal Waveform playback and Signal Capture)
C4	Trigger in (Signal Waveform playback and Signal Capture)
C5 – C7	Control connection to external units
System Ref Out	System reference output for Keysight accessories
10 MHz Ref In	Input connector for 10 MHz reference signal
10 MHz Ref Out	Output connector for 10 MHz reference signal
Power indicator	LED inside the stand-by button
Power LED	Power status: Green Power on
Status LED	System status: Green/off Status ok Red Warning (Indicated in case of overheating, over voltage or self-test failure. More detailed information about the reason of the warning is reported in the GUI.)
HDD LED	N/A
CH STATUS LED	Channel status (for each RF connector): No indication Connector not in use Green Status ok Blinking Emulation running Blue Connector selected in GUI Red Warning (Indicated in case of overheating, input cut-off, or self-test failure. More detailed information about the reason of the warning is reported in the GUI.)

1.5 Absolute maximum ratings

Absolute maximum ratings specify the maximum voltages and power levels. Exceeding the absolute maximum ratings may cause permanent damage for the unit. Correct operation is not guaranteed up to maximum values.

Table 2 Absolute maximum ratings

Port / connector	Absolute maximum rating
BRF1 RF IN/OUT	100 – 7250MHz: +27 dBm RMS / +36 dBm peak 3 – 100MHz: +7 dBm RMS / +16dBm peak

Port / connector	Absolute maximum rating
BRF1 RF OUT	100 – 7250MHz: +16 dBm peak (reverse) 3 – 100MHz: +10dBm peak (reverse)
BRF2 RF IN/OUT	>= 7250 MHz: +27 dBm RMS / +36 dBm peak
C1-C7	0 / +3.8 VDC
System Ref Out	+15 dBm 50 ohm (reverse)
10 MHz Ref In	+15 dBm 50 ohm
10 MHz Ref Out	+15 dBm 50 ohm (reverse)
AC INPUT connectors	264 VAC 50/60 Hz

1.6 Document history

The following table lists the main changes to issues of this document.

Issue	Date	Summary of Changes
1.1c	April 2024	BRF2 configurations added. Minor correction to Physical connectors and LEDs.
1.0	September 2022	PROPSIM FS16 (F8820B)

2 GETTING STARTED

This chapter explains how to get the PROPSIM up and running quickly after the emulator delivery.

2.1 Checking the delivery

Follow these steps in the initial inspection:

- Retain the order certificate upon product delivery.
- Verify that all contents have arrived as specified using the shipping lists attached to the delivery containers.
- Check that none of the shipping containers are damaged in any way. If a container has been damaged in shipping, verify that the emulator and all other delivered items are in an acceptable condition and that they function correctly, both mechanically and electrically.
- If something is wrong with the shipment, please contact Keysight. It is advisable to also record the incident by taking a photograph.
- Handle the delivery container with care and store it in a dry, safe place for later PROPSIM transportation purposes e.g. shipment for annual calibration.

Once the emulator and other parts have been checked, you can begin the actual installation process.

2.2 Installation

The PROPSIM emulator requires power source of 100 to 240 volts AC, single phase, 50/60 Hz, with a maximum of 1500 W for PROPSIM FS16 with full 16 channel / 32 RF ports configuration. Power cord must be connected behind separate 15 A fuse (minimum). The emulator should not be connected to AC power sources before verifying that the line voltages are correct. The emulator is provided with a three-wire power cord. When connected to appropriate AC power outlet, the cord grounds the emulator cabinet.

CAUTION
Power inlet of the device must be powered and switched on for the device operation. Power switch of the device must be off to ensure the device is powered off.

CAUTION
If you need to turn off the device in an **emergency**, unplug instrument mains or remove power cable at the rear of the instrument. For this reason, the power cable connection must be easily accessible – allowing turning off the device power quickly.

WARNING!
ELECTRIC SHOCK HAZARD! Before connecting the instrument to the power line or connecting the circuit that is connected to the power line, make sure that the protective earth terminal of the instrument is connected to the protective conductor of the power cord. Failure to do so could result in electrical shock.
Also, make sure that you insert the power plug into outlet that contains protective earth conductor (grounding). The grounding of one conductor on a two-conductor outlet is not sufficient for grounding purposes to prevent the possibility of electrical shock.

Use only the power cord delivered with the instrument.

Insert the power cord plug only into socket outlet provided with protective earth contact. Do not negate this protective action by using extension cord without protective conductor.

Allow the emulator to warm up to room temperature before turning the power on. Turning on a cold emulator may cause damage to the emulator due to possible water condensation.

IMPORTANT NOTE REGARDING THE INSTALLATION

When installing PROPSIM radio channel emulator, it is absolutely important to ensure adequate cooling of the device.

Ensure that the intended installation site has sufficient clearances for air flow around the emulator cabinet. The emulator requires unrestricted airflow for the fan and ventilation openings on the back and both sides. See Figure 5 below.

Make sure that there is at least 10cm (4”) of free space on both sides of the emulator.

The highest recommended ambient temperature is 25°C (77°F).

Figure 5 Air inflow and outflow on sides and back of the emulator

Perform the following hardware installation steps:

- Connect PROPSIM to an earth grounded main socket, minimum 15A fuse
- Connect a monitor (minimum display resolution 1024x768) to the DisplayPort connector.
- Connect a USB mouse and keyboard to the USB ports.

PROPSIM is delivered with Windows™ 10 IOT operating system and application software installed. No software installation actions are required by the user.

2.3 Startup

Perform the following steps to start up the PROPSIM:

- Turn on the main switch in rear panel.
- Press the Standby-button in the front panel to initiate emulator boot up.

The PROPSIM starts up and, by default, you are automatically logged in.

Your PROPSIM is now ready for use. Note however, that you should allow the emulator a **1-hour warm-up period** before doing measurements demanding high accuracy.

2.4 Shutdown

To shut down the PROPSIM, either press the Standby-button in the emulator front panel or select **System** > **Restart / Shutdown** > **Shutdown** in the navigation bar and click **OK** in the confirmation dialog, see Figure 6.

Note that keeping the Standby-button pressed down for five (5) seconds will result in uncontrolled shutdown of the application and operating system.

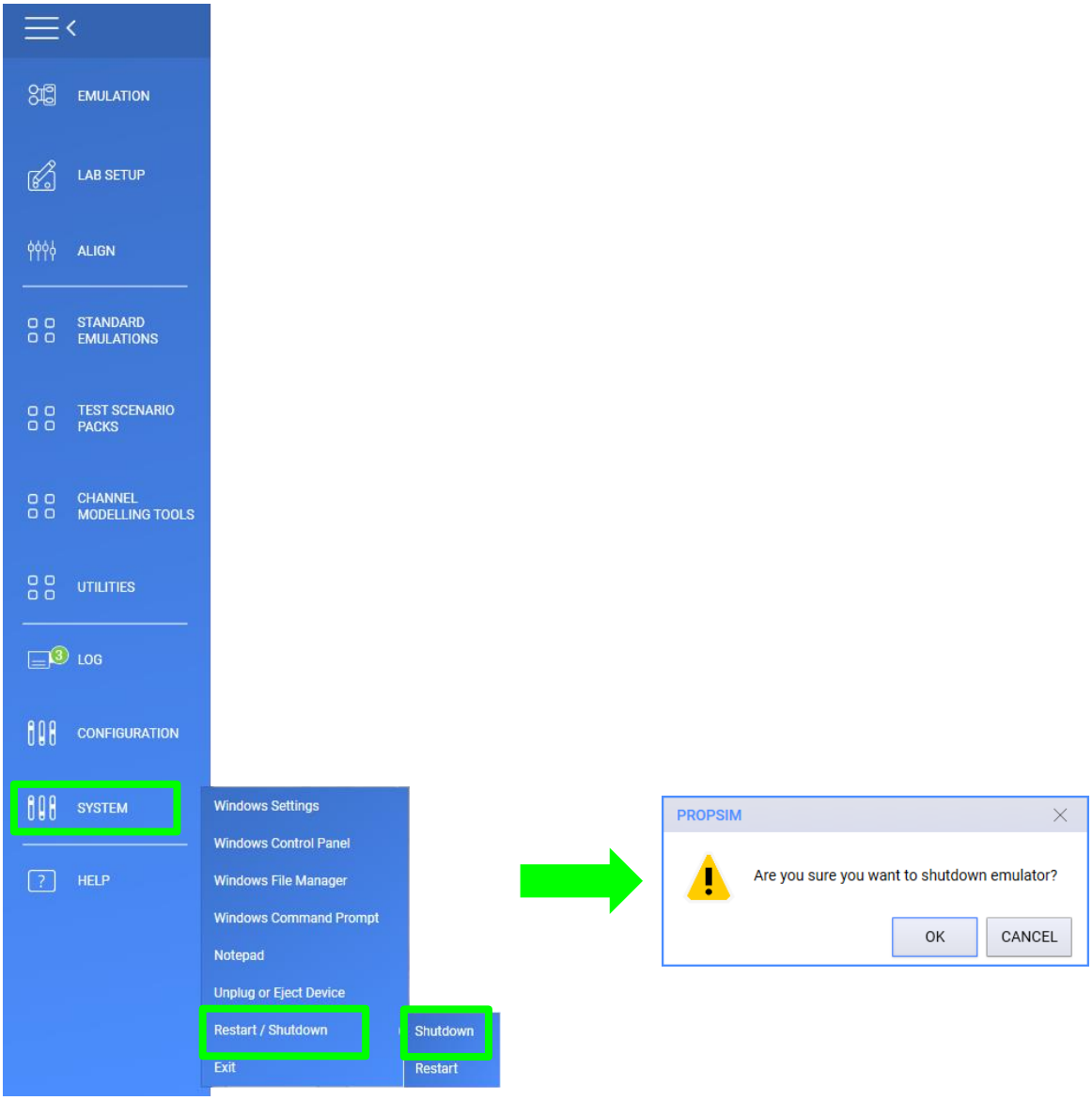


Figure 6 Shutdown from the Graphical User Interface

3 INTRODUCTION OF THE GUI

The Graphical User Interface (GUI) of the PROPSIM opens when the emulator has been started up and the user has logged in (see chapter 2.3).

Note: By default, the user is logged in automatically.

Figure 7 PROPSIM Home view

The GUI is divided into a navigation bar and a view area.

- The navigation bar is organized as a column in the left side of the GUI and provides easy access to all views and applications of the PROPSIM.
- The main views and applications of PROPSIM open in the view area. The content of the view area (including the menus and the tools available in the toolbar) depend on the selected view or application.
- The status of the system, reference clock, lab setup, and auto alignment is shown in the status bar at the bottom of the GUI.

Note: The navigation bar is not always shown in GUI screen shot examples.

3.1 Views, applications and tools

The PROPSIM includes the following views and applications:

- The **Home view** offers an easy access to pre-installed standard emulations, user defined emulations and **Scenario Wizard** to easily create emulations for different test scenarios, technologies and lab environments.
- The **Emulation control view** is used to run emulations pre-stored in the PROPSIM or created with the channel modelling tools. The Emulation control view also allows you to control many of the hardware emulation operations of the PROPSIM.
- The **Data views** offer different kinds of data about the emulation that is open or running.
- **Lab Setup** feature is used to store the common, typically fixed, laboratory and testing environment related PROPSIM parameters to separate 'lab setups'.
- **Auto Alignment** (optional feature) is a tool for aligning phases and levels between multiple channels in the test setup.
- **Test scenario packs** contain additional, separately installed, ready-made emulations and corresponding documentation for testing of technology or application specific standardized scenarios.

- Utilities:
  - Channel model view is used to create statistical channel models.
  - Correlation Editor is used to define statistical correlation between channels of correlative or MIMO models.
  - ASO editor is a tool for creating aerospace channel models.
  - Run-time Model Editor is used to create models for emulations using run-time control.
  - Multi Emulator Scaler is a tool for aligning emulation gains/levels in multi-emulator setups. This tool is internally used by channel modelling tools and used separately only in special cases.
  - Batch Builder is a tool for building all the selected emulations at once.
  - Running view is a legacy view that can be used to load and run emulations.
- Data logging can be used to store emulation run-time data, such as power levels, speeds, etc. to a local file or stream it out with LAN UDP protocol.
- Shadowing editor (optional feature) is used for defining shadowing profiles for channels. It can also be used to generate path loss triggered handovers / handoffs. The Shadowing editor is part of Scenario Wizard.
- Interference generator (optional feature) is used to create interference sources for the emulations. Interferer can be added to an emulation in Scenario Wizard or Running view.
- Geometric Channel Modeling (GCM) Tool (optional feature) is a channel modeling tool for creating dynamic spatial channel models.
- VDT RF field to lab (optional feature) Automated, lab-based performance and interoperability test solution for mobile devices and network infrastructure.
- WLAN Modeling Tool (optional feature) is used to generate radio channel realizations according to IEEE 802.11n/ac/ax Channel Models specification.

The following chapters explain in detail how to use these views for and how the channel models from different applications can be used.

3.2 Navigation bar

The navigation bar provides access to different views, applications and tools. The structure of the navigation bar is described in Table 3.

Table 3 Navigation bar items

Menu Item	Submenu Item	Description
Home		Return to the home view from other views (such as Lab Setup). ***Note:*** *When an emulation is open in the background, clicking the Home button opens the Emulation Control View instead of the Home view.*
Save		Save changes in the Emulation Control View.
Expand/Collapse Navigation Bar		Show/hide navigation bar menu items.
Emulation	New	Create a new emulation in the Scenario Wizard.
	Open	Open an existing emulation in the Emulation Control View.
	Save	Save changes in the emulation that is open in the Emulation Control View.
	Close	Close the current emulation.
	Close and edit	Close the current emulation and open it to Scenario Wizard.

Menu Item	Submenu Item	Description
Lab Setup	New	Create a new lab setup in the lab setup editor. The lab setup editor is used to configure operation parameters that can be repeated for several emulations.
	Open	Open an existing lab setup for editing in the lab setup editor.
	Export	Export lab setup from the currently open emulation.
	Save	Save changes in the lab setup that is open in the lab setup editor.
	Deselect	Take the current lab setup out of use.
	Exit	Close the lab setup editor.
Align	New Auto Alignment	Create a new auto alignment.
	Open Auto Alignment	Select an existing auto alignment file for use.
	Deselect Auto Alignment	Take the current auto alignment out of use.
	New User Alignment	Create a new user alignment.
	Open User Alignment	Select an existing user alignment file for use.
	Deselect User Alignment	Take the current user alignment out of use.
Standard Emulations		Access to pre-installed standard emulations.
Test Scenario Packs		Access to test scenario packs. Test scenario packs contain additional, separately installed, ready-made emulations and corresponding documentation for testing of technology or application specific standardized scenarios.
Channel Modeling Tools		Access to channel modeling tools.
Utilities	Channel Model View	Open the channel modelling tool.
	Correlation Editor	Open the tool for creating correlation matrices.
	ASO Editor	Open the aerospace channel model editor.
	Runtime Model Editor	Open the runtime model editor.
	Multi Emulator Scaler	Open the multi emulator scaler.
	Batch Builder	Open the emulation batch builder.
	IR and ASC Converter	Open the tool for converting between ASC and IR -files.
	PROPSIM Backup Tool	Opens the tool for taking backups of, for example, user data, lab setups, auto alignments, licenses, etc.
	Running View (legacy)	Open the running view (legacy).
Log		Open the system log. You can also set the log to show a notification callout for warnings and errors.
Configuration	Device Information	Includes, for example, serial number and IP address.
	Device Configuration	Includes information about emulator hardware and option configuration
	Import License	Open a dialog for importing and/or updating licenses, see chapter 2.13

Menu Item	Submenu Item	Description
	External RF unit configuration	Open a dialog for selecting configuration for external RF units (up/down -converters). Requires options F8800AUD1/ M1742A (RRH's), F8800ACI1(CIU)/S9165A or F8800AMP1.
	External unit manager	Opens external unit manager for creating configurations for external RF units. Requires options F8800AUD1/ M1742A (RRH's), F8800ACI1(CIU)/S9165A or F8800AMP1.
	System Backup / Restore	Opens the tool for taking backups of, for example, user data, lab setups, auto alignments, licenses, etc.
System	Windows Settings	Configuration of several emulator details, such as locale, time zone and network via Windows Settings dialog.
	Windows Control Panel	Configuration of several emulator details, such as locale, time zone and network via Windows Control Panel.
	Windows File Manager	Can be used for browsing, deleting or moving emulations.
	Windows Command Prompt	Text-based file- and control utility.
	Notepad	For small note keeping.
	Unplug or Eject Device	Utility to safely unplug removable devices, such as USB mass memories.
	Restart / Shutdown	Controlled restart and shutdown of emulator.
	Exit	Controlled exit of the PROPSIM software.
Help	Documents	Quick Start Guide, User Reference, and Application Note documents in PDF format.
	About	The About dialog includes firmware release information.
Help – Technical Support	My Software Support	Open My Software Support view containing details of licenses configuration including support and license expiration date information.
	My Support ID	Show device type and Support ID (serial number).
	Export system log files	Open a dialog for exporting the system logs as text files in a zipped folder.

3.3 Status bar

The status of the system, reference clock, lab setup, and auto alignment is indicated with green, orange, and grey colours in the status bar as shown in Table 4.

Table 4 System status items

Category	Item	Colour	Description
Status			Main level system status
	STATUS	Green	No errors or warnings
	STATUS	Orange	System errors.
	STATUS	Red	System errors.
Reference Clock			

Category	Item	Colour	Description
	INT REF	Green	Internal reference clock selected.
	INT REF	Orange	Internal reference clock warming up.
	EXT REF	Green	External reference clock selected and OK.
	EXT REF	Orange	Locking error to external reference clock. ***Note:*** *Locking might take a few seconds.*
Lab Setup			
	LAB	Grey	No lab setup selected, lab setup is not in use.
	LAB	Green	Lab setup selected and in use.
	LAB	Orange	Lab setup selected, but cannot be used with current emulation.
	LAB	Red	Error in selected lab setup file
Auto Alignment			(Optional feature)
	ALIGN	Grey	No auto alignment file selected, auto alignment is not in use.
	ALIGN	Green	Auto alignment file selected and in use.
	ALIGN	Orange	Auto alignment file selected, old alignment.
	ALIGN	Red	Auto alignment has expired.

Clicking the STATUS button opens the System log dialog shown in Figure 8. The system log collects system event and alarms, reference clock changes, etc.

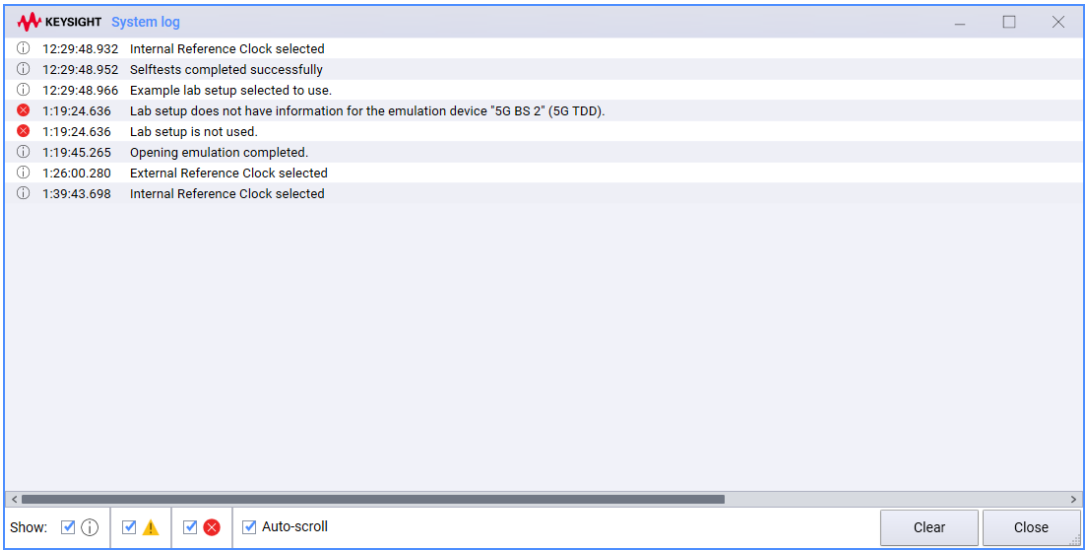


Figure 8 System log dialog

Instructions for collecting system information when contacting helpdesk can be found from Troubleshooting section in User Reference documentation.

4 BASIC FUNCTIONS

The Home view of the PROPSIM GUI that was presented in Figure 7 (chapter 3) includes an easy to use **Scenario Wizard** (see Figure 9 and chapter 4.4) for creating new test scenarios for different technologies. The Scenario Wizard provides easy steps from configuring the basic settings to finalizing the test setup. It includes several radio technology definitions that pre-define common environment variables (e.g. crest factor, maximum transmit power, center frequency) for your test scenario.

The Home view also offers you a quick access to pre-installed standard emulations, user emulations and recently used emulations.

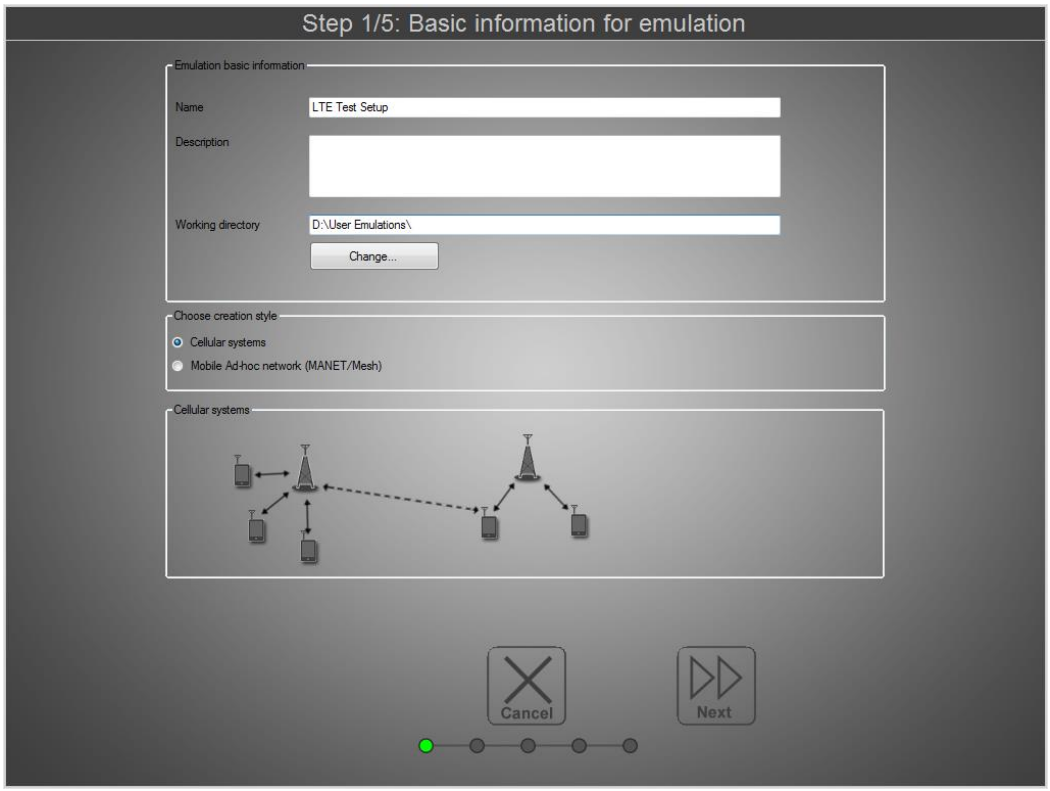


Figure 9 Scenario Wizard. The example illustrates first step of creating emulation for Cellular systems

The **Emulation control view** of the PROPSIM GUI (see Figure 10) is used to run emulations pre-stored in the PROPSIM or created with the channel modeling tools. The Emulation control view also allows you to control many of the hardware emulation operations of the PROPSIM and save changes to the emulation for future use.

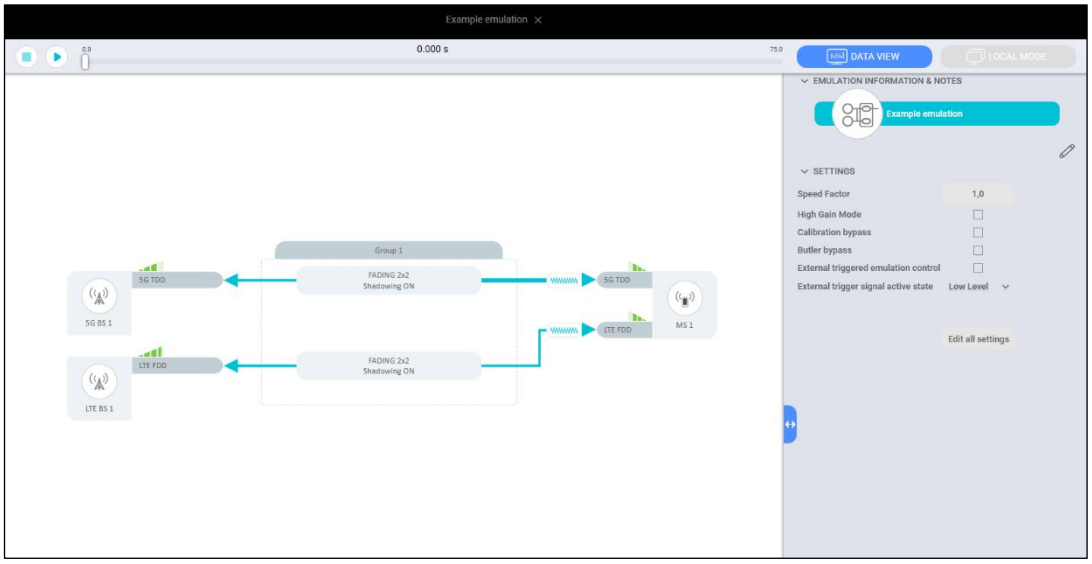


Figure 10 Emulation control view. The example illustrates test scenario with two base stations and one mobile station

The emulation setup is presented as a diagram in the Emulation control view. The emulation diagram consists of base stations and mobile stations with their technologies, groups, fading elements, links, and optional interferences. When you select an element in the diagram, the settings of that element are shown in the settings pane on the right. When none of the elements are selected, the settings pane shows the emulation settings.

4.1 Loading emulations

To open an emulation in the emulation control view, click the **Edit or Run emulation button** in the Home view, or select **Emulation > Open** in the Navigation bar.

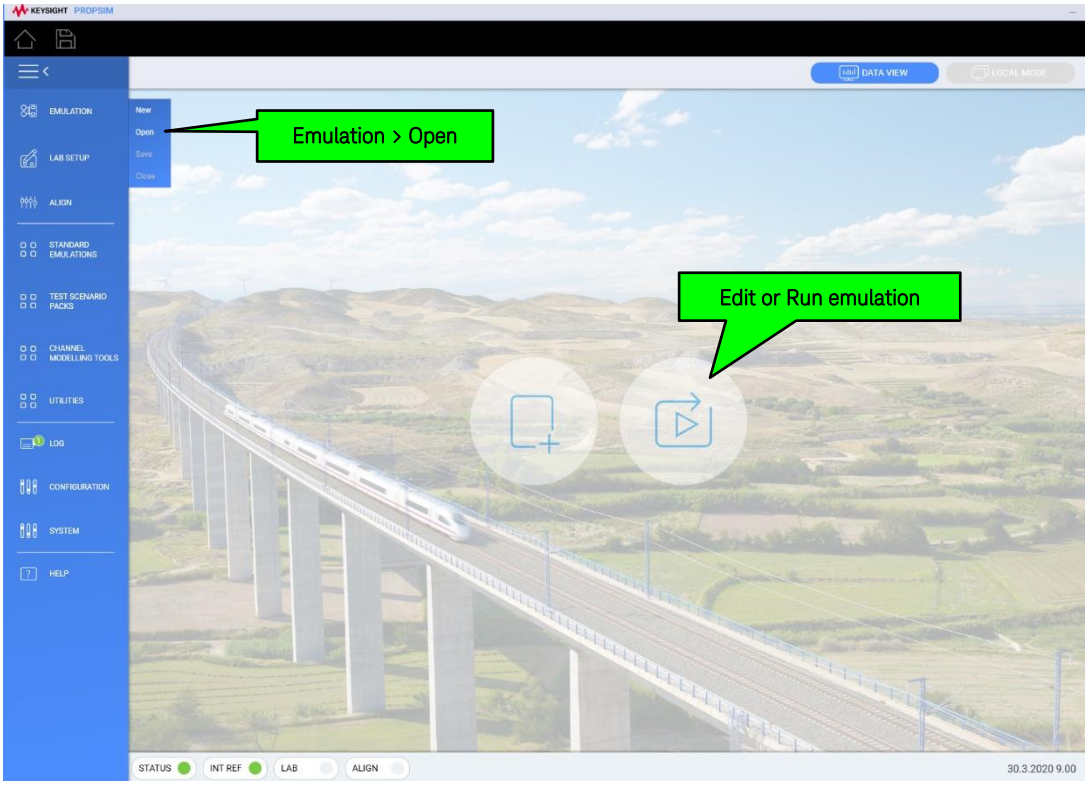


Figure 11 Open emulation

The **Open Emulation** dialog appears (see Figure 12). The PROPSIM includes pre-stored standard emulations which enable quick emulator setup for standard test cases. Standard emulations are listed in hierarchical structure in **All emulations** list at the right side of the dialog, along with user emulations. The recently used emulations can be quickly accessed at the left side menu of the dialog.

Select the emulation (SMU file) that you want to run and click **Run**.

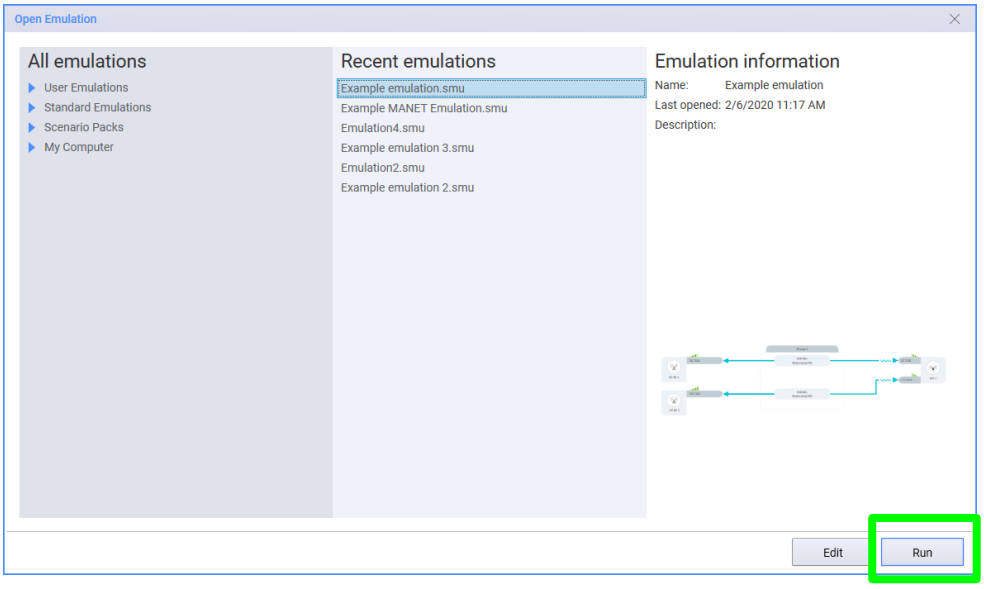


Figure 12 Selecting an emulation to run

The emulation diagram defined in the SMU file opens in the Emulation control view.

After a selected emulation is loaded to the Emulation control view, the input and output power levels, center frequency and mobile speed parameters can be modified.

4.2 Running emulations

The emulation timeline in the Emulation control view contains the main controls for running an emulation.

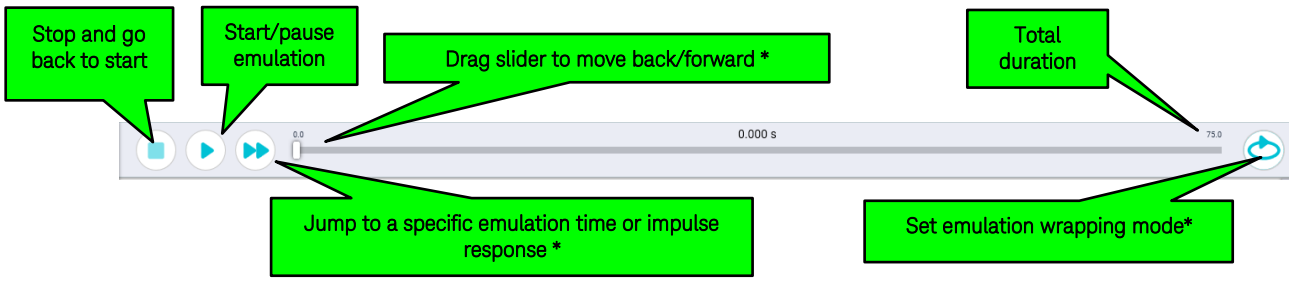


Figure 13 Emulation timeline *(* You cannot move to another time on the timeline when the emulation is running.)*

The **Start** button starts the channel emulation process. By default, the emulation runs continuously: it will loop back to the beginning when it has reached the end of the channel emulation. **Wrapping mode** can be configured with the button on the right.

To stop an emulation run, click the **Stop** button . This will set the emulation back to its initial state, meaning back to the first IR (Impulse Response). If you click **Start** after stopping the emulation, the emulation starts from the beginning.

To pause the emulation, click the **Pause** button . This pauses the emulation to the IR that was active when the button was clicked. If you click **Start** again, the emulation continues from the IR where it was paused.

While the emulation is stopped or paused, you can move the emulation back and forward on the timeline by dragging the slider. You can also jump to a specific emulation time or impulse response by clicking the emulation running time and entering a time or impulse response.

4.3 Viewing emulation data

The PROPSIM data views offer different kinds of data about the emulation that is open or running:

- Graph with measured level, shadowing, and interference curves of the links
- Live numeric data of antenna input/output powers and link dopplers
- Connectors on the emulator front panel
- Port RF level graph
- Channel impulse response
- Aerospace link information and graphs
- Map view for GCM models
- ATE LAN monitor

To open the Data views window, click the **Data view** button on the top right corner of the PROPSIM software.

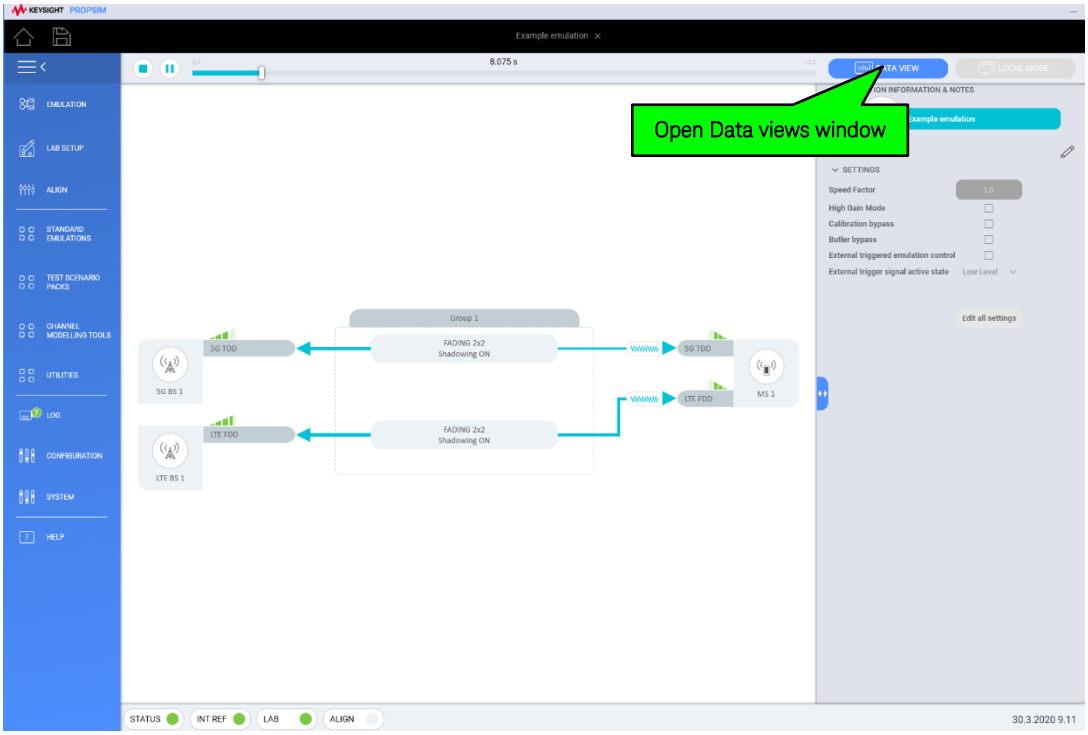


Figure 14 Opening Data view

4.3.1 Viewing IR changes

You can view the change of the IRs during emulation in the **CIR** (Channel Impulse Response) view in the Data views window. The CIR view shows the change of the channel input responses (CIRs) during the emulation. The signal paths of the channel model are visualized in the CIR graph as a power delay profile. Relative and absolute delay and gain values for each path can be seen in a tooltip.

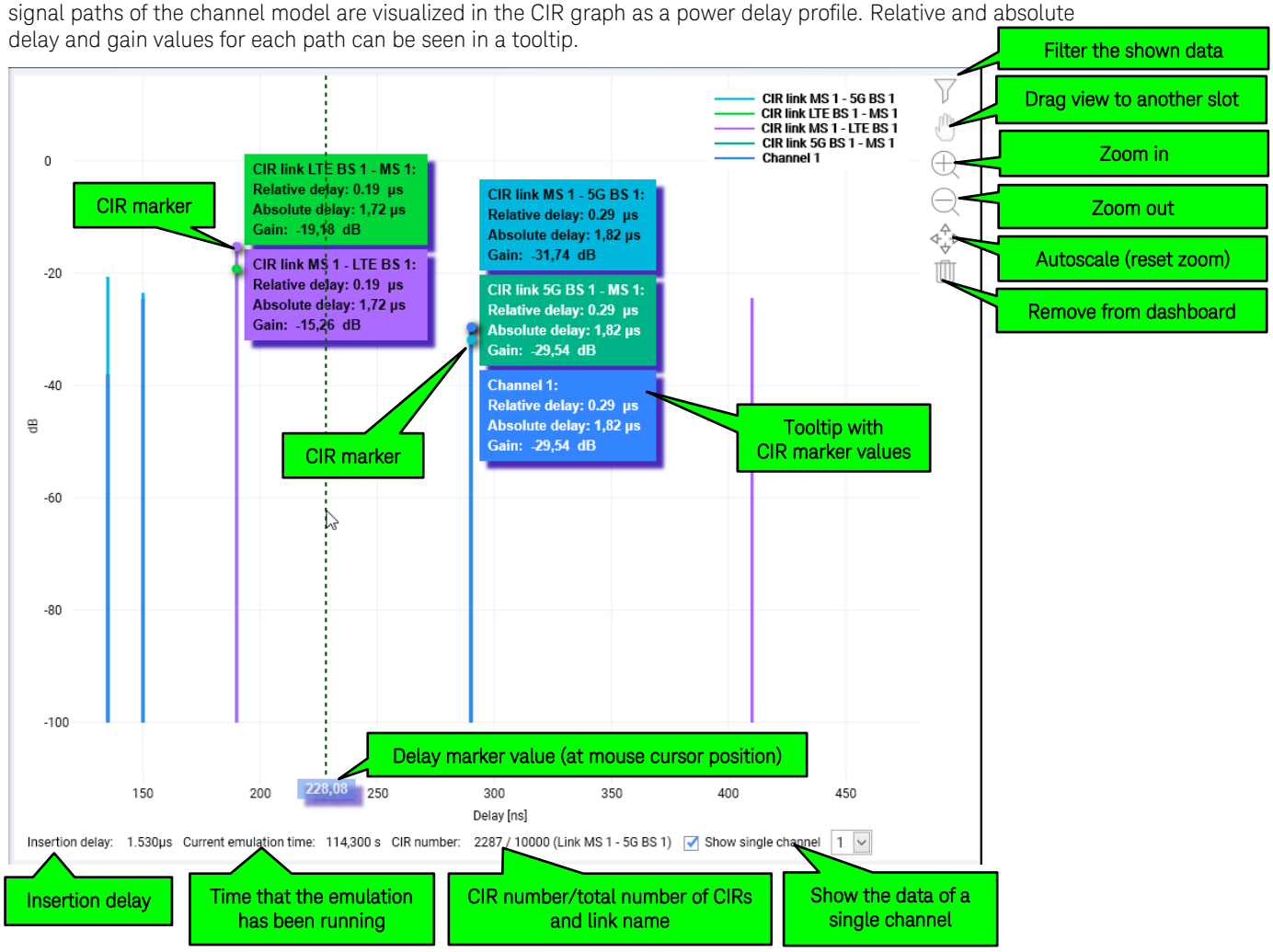


Figure 15 CIR view

To select which links are shown on the graph, click the (Filter) button in the data view toolbar.

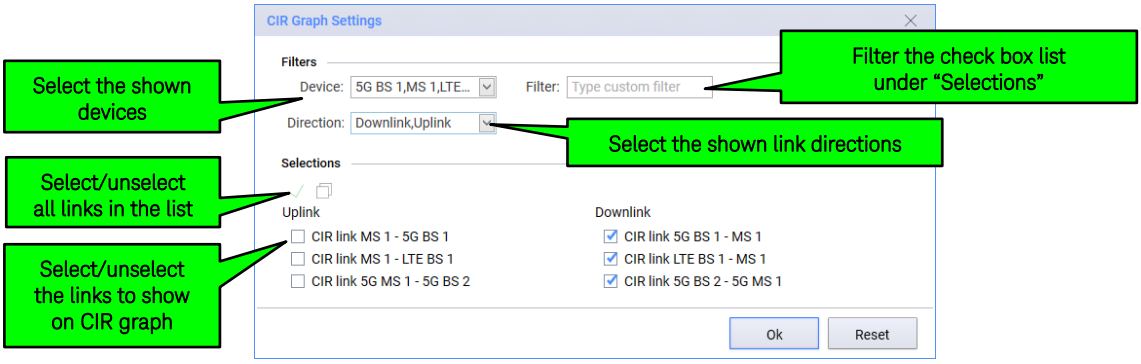


Figure 16 CIR Graph Settings dialog

In addition to links, you can also view the CIR data of a single channel by selecting the **Show single channel** option and the channel you want to see in the drop-down list.

Insertion delay describes the internal hardware delay with the used channel model. The total real delay of a tap is the sum of the insertion delay and the delay value defined in the channel model. If the smallest delay in the channel model is greater than the insertion delay, the insertion delay is zero.

4.4 Creating emulations with Scenario Wizard

Emulation creation for running with Scenario Wizard is a five-step process, as illustrated in Figure 17.

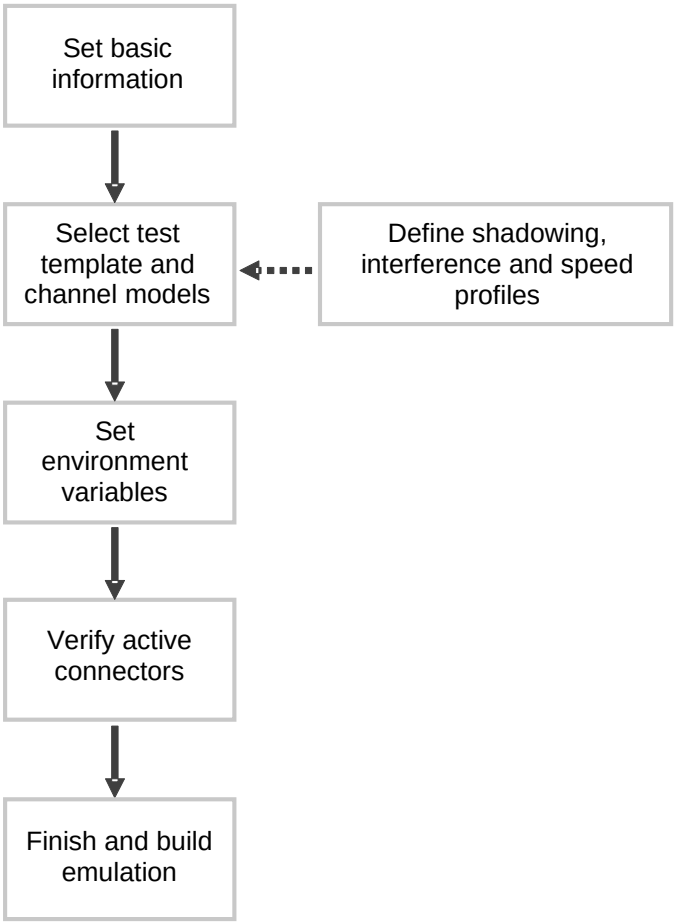


Figure 17 Emulation creation process in Scenario wizard

Scenario Wizard has a separate page for each step in the creation process. First, you must define the basic information for new emulation on the first page. Then, on the next page, you set the appropriate pre-defined channel models either by browsing one for all logical channels or by defining separate channel models for each channel between the transmitter and receiver antennas. It is also possible to create and browse shadowing profiles for link outputs (optional). After the channel model settings are completed, you can modify the technology based pre-defined environment variables (for device based, MIMO, SIMO, SISO and MISO emulations) and verify the active connector setup.

When all the steps above have been completed, emulation can be saved and built for running it in the Emulation control view.

Scenario wizard contains navigation buttons (presented in Figure 18) for navigating between the wizard steps (Back and Next) and for cancelling the wizard changes (Cancel). The current step is indicated with green color under the buttons.

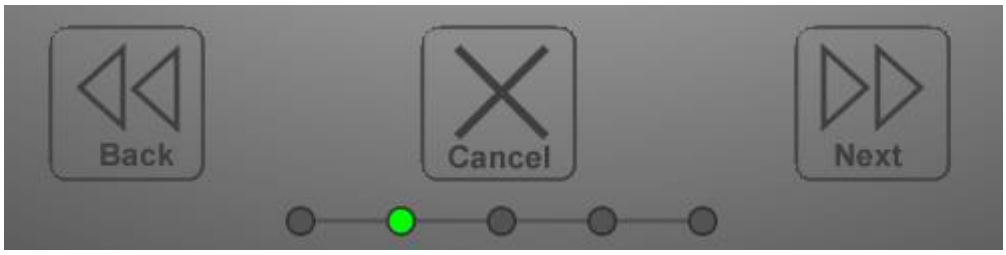


Figure 18 Scenario Wizard navigation

To launch the Scenario wizard, click the New emulation button in the Home view.

4.4.1 Creating new bi-directional MIMO channel emulation

Figure 19 illustrates an LTE FDD test scenario example with a single base station and a single mobile both with two antennas for data throughput testing over bi-directional MIMO link.

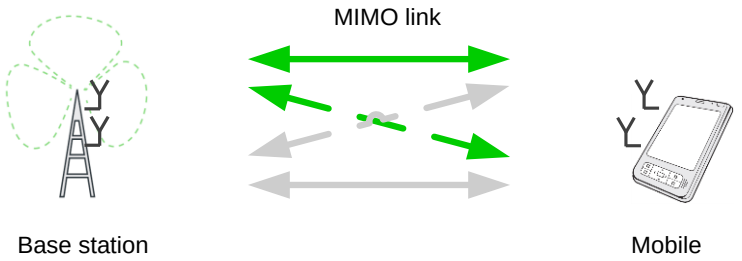


Figure 19 Bi-directional MIMO link

This section explains how to create bi-directional 2x2 MIMO channel emulation with Scenario Wizard. The test scenario illustrated in Figure 19 is used as a baseline for this example.

Note: To run bi-directional FDD 2x2 MIMO emulation in the PROPSIM, two RF locals (downlink and uplink are using different frequencies in FDD) and at least four RF channels are required.

4.4.1.1 Step 1: Set basic information for emulation

On the first page of wizard, you can modify the name and add a description for your test scenario (see Figure 20). Scenario wizard creates emulations to user emulations folder. You can browse the default working folder or make needed folders in it.

Figure 20 Emulation basic information

Select the needed bandwidth (see Figure 21).

Figure 21 Bandwidth selection

Bandwidth options available depend on the PROPSIM channel bandwidth configuration and installed MIMO Extension option license of PROPSIM.

Next, select **Cellular systems** option as your emulation creation style (see Figure 22) and click the **Next** button in the wizard navigation bar to proceed to the next page.

Figure 22 Emulation configuration selection

4.4.1.2 Step 2a: Select test setup template

Next step is to define the appropriate test setup for your emulation. By default, a template for “Single link” (one BS and one MS) is selected and it is used in this example, as shown in Figure 23.

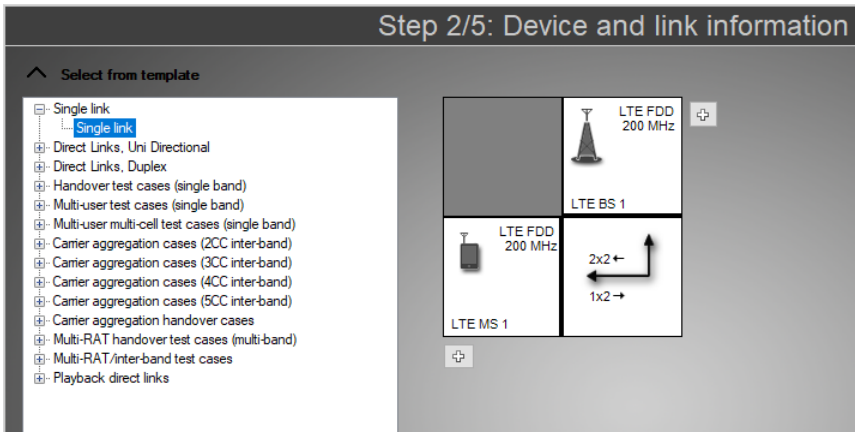


Figure 23 “Single link” selected as test setup template

4.4.1.3 Step 2b: Configure device technologies and antennas

Click MS in the figure to open the device properties, as shown in Figure 24.

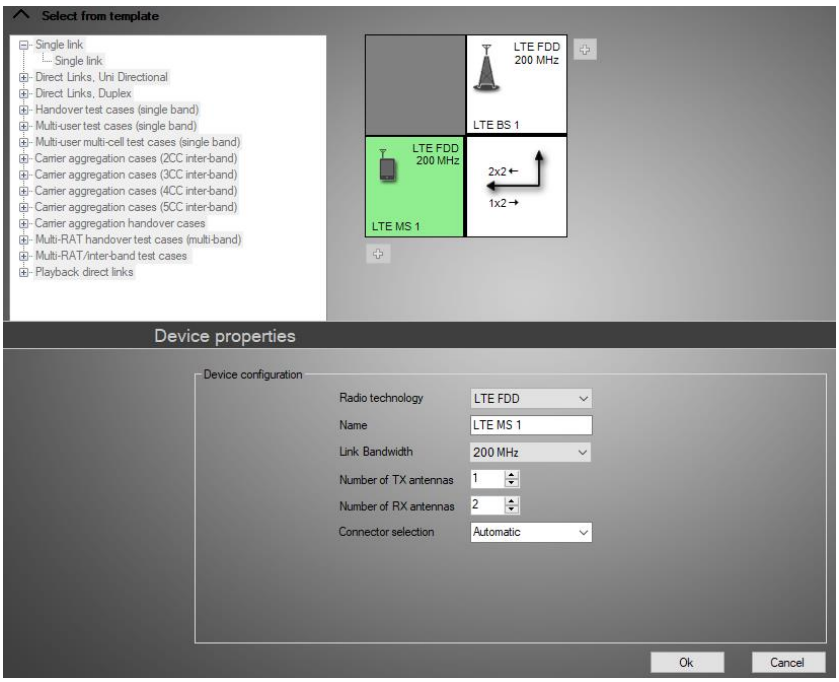


Figure 24 Device (MS) properties

Select the desired radio technology in the **Radio Technology** drop-down list. This selection pre-defines common environment variables in your test scenario (e.g. crest factor, maximum transmit power). Here we select LTE FDD. **Number of TX and RX antennas** defines the MIMO dimensions for uplink and downlink. This example uses 1 TX and 2 RX antennas for mobile station, and correspondingly, 2 RX and 2 TX for the base station. This results in 2x2 MIMO for downlink and 2x1 MISO for uplink. Click **OK** to close the device properties.

4.4.1.4 Step 2c: Configure channel models

Click the link between MS and BS to open the link properties page shown in Figure 25.

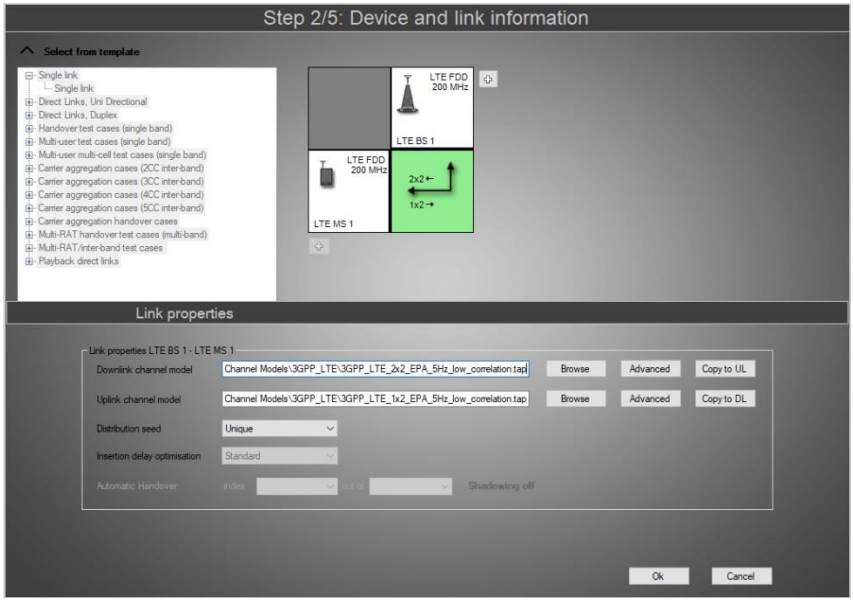


Figure 25 Link properties page

Default channel models are selected automatically from the installed Standard Channel Models package, according to selected radio technology and antenna configuration.

To change the model for uplink or downlink, click **Browse** to select appropriate channel models for your emulation, see Figure 26.

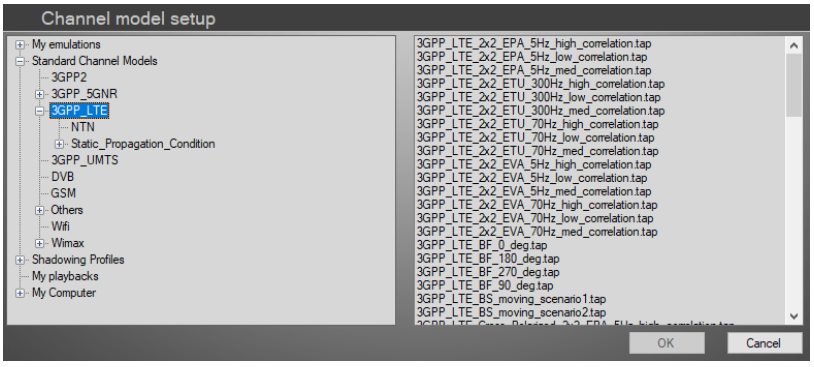


Figure 26 Selecting channel model

The folders are listed on the left side of the window, and the channel model files on the right side. When you select a folder, the channel model files that fit to the number of your link antennas are listed in the file list.

For this LTE FDD example, we keep the default **3GPP_LTE_2x2_EPA_5Hz_low_correlation.tap** file from **3GPP_LTE** folder for downlink, and the default **3GPP_LTE_1x2_EPA_5Hz_low_correlation.tap file** for the uplink.

The distribution seed for channel model creation random process can be defined for each link. The uplink and the downlink always use the same seed.

When you have selected the channel models for both links, click **OK** to close the link properties. Then, click **Next** to proceed to the next page.

4.4.1.5 Step 3: Configure environment variables

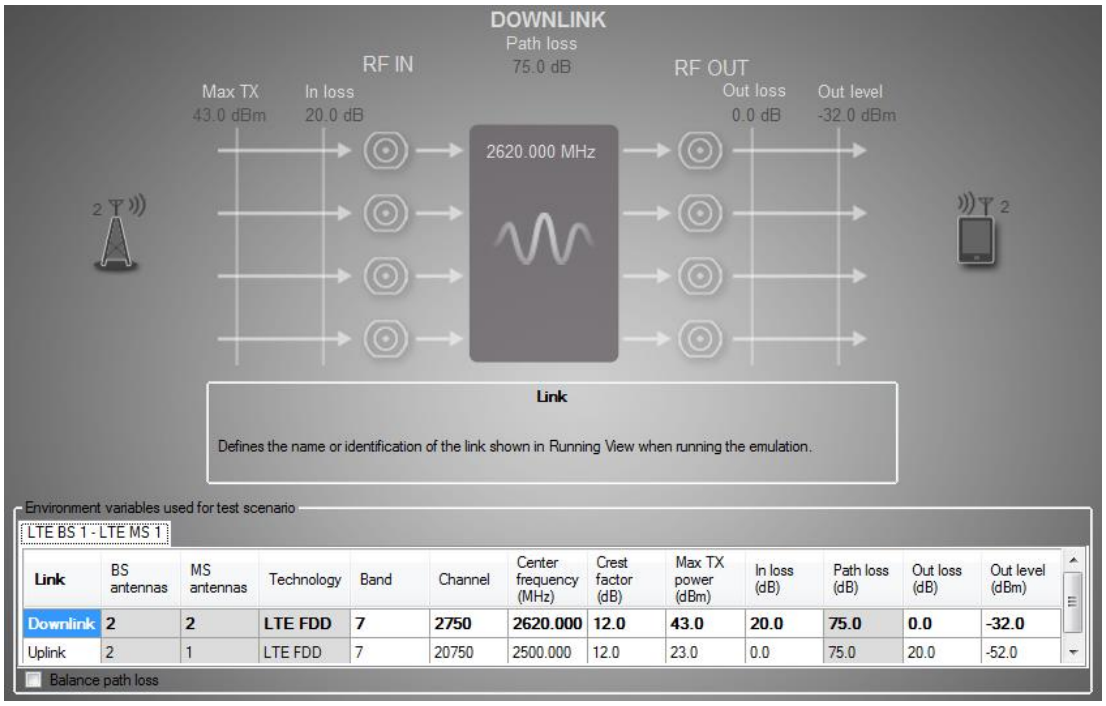
The next step is to define and configure your test environment variables for the emulation.

Let's use the following test environment in this example test scenario:

- Center frequency for the downlink is **2160 MHz**, and for the uplink **1970 MHz**
- Maximum RF (RMS) power of the base station is **+39 dBm,** and the mobile **+23 dBm**
- The signal Crest factor for both the base station and the mobile is **+15 dB**
- Actual path loss of the cable and other components from the base station to the emulator input is **+31 dB**

- Actual path loss from the emulator output to mobile is **+8 dB**
- The total target path loss for both directions is **102 dB**

Figure 27 presents predefined environment variables for bi-directional 2x2 MIMO channel emulation. Predefined values are based on the selected technology (in this example, LTE FDD).



Link	BS antennas	MS antennas	Technology	Band	Channel	Center frequency (MHz)	Crest factor (dB)	Max TX power (dBm)	In loss (dB)	Path loss (dB)	Out loss (dB)	Out level (dBm)
Downlink	**2**	**2**	**LTE FDD**	**7**	**2750**	**2620.000**	**12.0**	**43.0**	**20.0**	**75.0**	**0.0**	**-32.0**
Uplink	2	1	LTE FDD	7	20750	2500.000	12.0	23.0	0.0	75.0	20.0	-52.0

Figure 27 Predefined emulation environment variables

In Figure 27, the selected link in the view is Downlink. The values of the selected link are also illustrated.

To setup the example test environment, follow these steps:

- First, change Downlink center frequency to 2160 MHz, and Uplink center frequency to 1970 MHz
- Change Downlink Max TX Power to +39 dBm, and Uplink Max TX Power to +23 dBm.
- The downlink In loss is defined to be 20 dB. However, the path loss from the base station to the emulator input has been measured to 31 dB. Change that value to downlink In loss field.
  - Note: In bi-directional channel emulations, when you change the In loss value of the downlink, the Out loss for uplink changes similarly and vice versa.
- The downlink Out loss is defined to be 0 dB, change it to 8 dB.

Now the environment variables should look like presented in Figure 28:

Downlink	**2**	**2**	**LTE FDD**	**1**	**500**	**2160.000**	**15.0**	**39.0**	**31.0**	**82.0**	**8.0**	**-43.0**
Uplink	2	1	LTE FDD	1	18500	1970.000	15.0	23.0	8.0	89.0	31.0	-66.0

Figure 28 Environment variables after In loss change.

- The defined total system path loss target is 102 dB. The path loss is calculated:
  - Path loss = Base station Tx level - Mobile Rx level = 39 dBm - (-43 dBm) => 82 dB
- Output level of downlink must be reduced from -43 dB to -63 dB to get the desired 102 dB downlink path loss
- Similarly, output level of uplink must be reduced from -66 dB to -79 to get the desired 102 dB uplink path loss

Now all parameters should look as presented in Figure 29:

Environment variables used for test scenario

LTE BS 1 - LTE MS 1

Link	BS antennas	MS antennas	Technology	Band	Channel	Center frequency (MHz)	Crest factor (dB)	Max TX power (dBm)	In loss (dB)	Path loss (dB)	Out loss (dB)	Out level (dBm)
Downlink	2	2	LTE FDD	1	500	2160.000	15.0	39.0	31.0	102.0	8.0	-63.0
Uplink	2	1	LTE FDD	1	18500	1970.000	15.0	23.0	8.0	102.0	31.0	-79.0

Balance path loss

Figure 29 Finished environment variables

Now all environment variables have been set. Press **Next** to proceed to next page.

4.4.1.6 Step 4: Verify active connectors

Scenario Wizard automatically allocates physical RF connectors that are used when running the emulation. Allocation is based on selections in previous steps. On this page, you can verify the default connector settings or modify the connector locations to match test setup cable connections.

Figure 30 Active connectors selection

Figure 30 presents the PROPSIM F16 default active connectors for bi-directional 2x2 MIMO channel emulation example, as located in the front panel.

If you want to change the default connector setup, you can do it in two ways:

- Clear active connectors by clicking the **Clear All** button, and then drag all antennas one by one to the RF connectors you want to use.
- Drag and drop the antennas to the RF connectors you want to use. When dragging an antenna to a new connector, a suitable connector is shown as green in the front panel:

You can restore the default connector setup by clicking the **Default** button.

4.4.1.7 Step 4: Finish and build emulation

The last page of the wizard presents a brief summary of selected emulation configuration (see Figure 31).

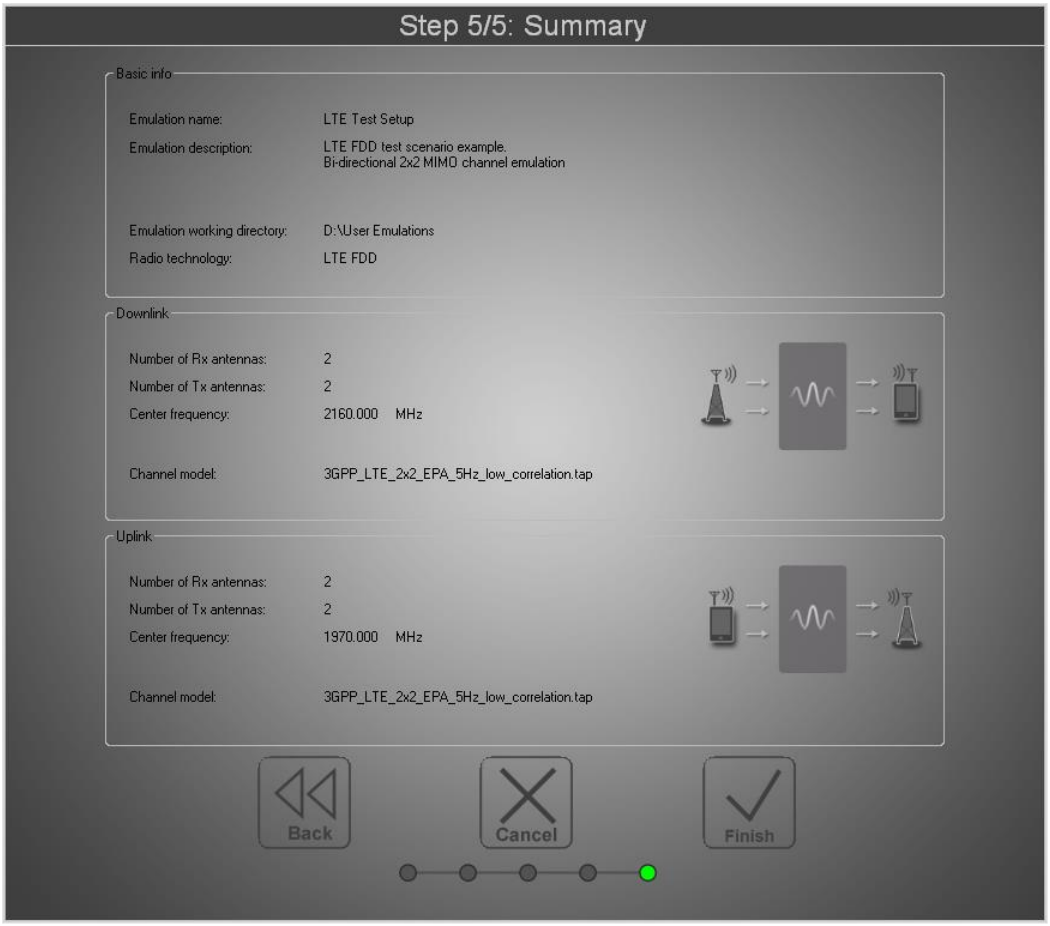


Figure 31 Emulation summary

In this page, you have option to go back to modify previous selections or cancel all changes (emulation will not be saved). To create emulation, click the **Finish** button.

When **Finish** emulation button has been clicked, you will have four options:

- **Build & Finish emulation**: Save emulation and build it.
- **Build & Run emulation**: Save emulation, build it and open to the Emulation Control View.
- **Finish emulation**: Save emulation.
- **Cancel**: Cancel and return to summary page.

To finish this example, select **Build & Finish Emulation** to save and generate an emulation that will be ready for running. Build progress window appears to the bottom of the window (see Figure 32).

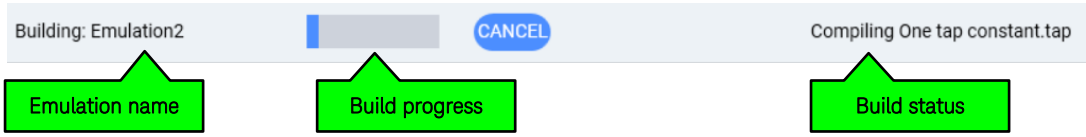


Figure 32 Build progress bar

At the left side of the progress window, you can see the name of the emulation. Progress is indicated with blue color in progress bar, and build status is shown at the right side of the progress window. To get more specific build messages, hover the mouse over the build progress bar.

When the build has finished, the emulation is ready for running.

4.4.2 Creating new handover MIMO channel emulation

This section explains how to create handover MIMO channel emulation with two base stations and one mobile illustrated in Figure 33.

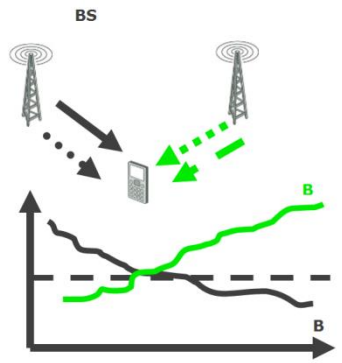


Figure 33 Path loss handover

The handover emulation is created with Scenario Wizard, using a provided handover template. To launch the Scenario wizard, click the **New emulation** button in the **Home view**.

4.4.2.1 Step 1: Set basic information for emulation

In first page of the wizard, the basic emulation information is defined the same way as in the previous chapter 4.4.1.1.

Select **Cellular systems** as the creation style as shown in Figure 34.

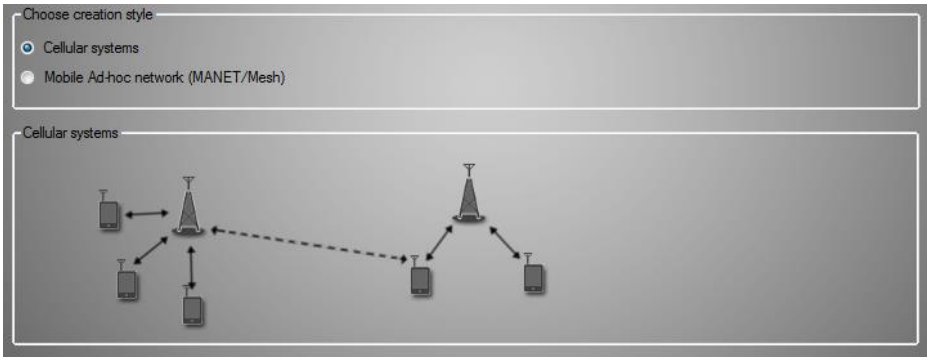


Figure 34 Creating new handover emulation

After finishing the basic settings, click the **Next** button in the wizard navigation bar to proceed to the next page.

4.4.2.2 Step 2: Select template and channel models

The multiple links setup page is illustrated in Figure 35. The left side of the page contains a list of available scenario templates where you can select a suitable configuration for your test setup. The right side of the page shows the devices and links of the selected scenario. For this example, open the category "**Multi-RAT handover test cases (multi-band)**" and double-click "**2 Base stations**".

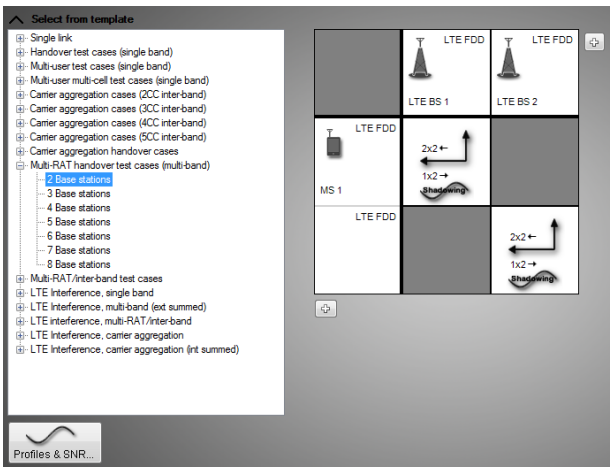


Figure 35 Selecting handover template with two base stations

The default fast fading models are selected automatically according to the selected technologies and antenna configurations. If you want, you can change the fast fading channel models for each link by clicking any "link" element and changing the models there, as discussed in chapter 0. The default path-loss curve for the handover test is automatically applied (duration 15 s, 30 dB attenuation ramp).

If needed, you can modify the hand-over shadowing profiles by clicking "**Profiles & SNR...**" as shown in Figure 36. To add additional lognormal variation, select the "**Apply lognormal**" checkbox.

Note: Right clicking the "device" or "link" element opens a context menu for editing shadowing or interference of the clicked element.

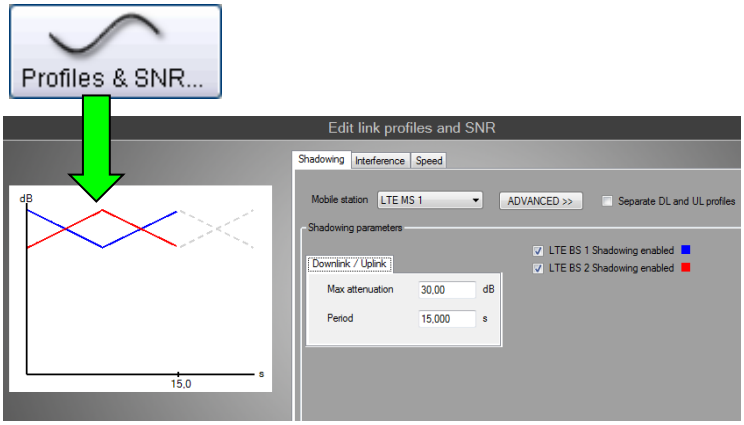


Figure 36 Modifying hand-over shadowing parameters

Click **Next** to proceed to environment variables page.

4.4.2.3 Step 3: Configure environment variables

When creating an emulation with multiple links - as now in the handover scenario – the environment variables of the links are on separate sheets as shown in Figure 37 and can therefore be configured independently. The parameter setup is identical to the previous example, see chapter 4.4.1.5.

Note: When using "Balance path loss" option, remember to enable it for all links separately.

Environment variables used for test scenario

BS 1 - MS 1 | BS 2 - MS 1

Link	BS antennas	MS antennas	Technology	Band	Channel	Center frequency (MHz)	Crest factor (dB)	Max TX power (dBm)	In loss (dB)	Path loss (dB)	Out loss (dB)	Out level (dBm)
Downlink	**2**	**2**	**LTE FDD**	**7**	**2750**	**2620.000**	**15.0**	**43.0**	**28.0**	**106.0**	**8.0**	**-63.0**
Uplink	2	1	LTE FDD	7	20750	2500.000	15.0	23.0	8.0	106.0	28.0	-83.0

☐ Balance path loss

Figure 37 Environment parameters for both links (MS1 - BS1 and MS1 - BS2)

Click **Next** to proceed to the connector setup page.

4.4.2.4 Step 4: Verify active connectors

The default connector setup is presented in Figure 38. For clarity, all the connections in the figure are visible simultaneously and highlighted with green and blue colors.

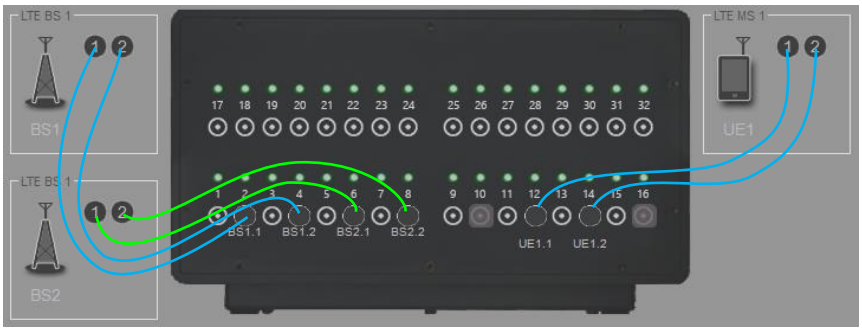


Figure 38 Default connector setup of 2 base stations handover scenario

As shown in this multi-band scenario, PROPSIM F16 can combine different frequency bands into UE antenna connectors, without using external combiners.

Click **Next** to proceed to the summary page.

4.4.2.5 Step 5: Finish and build emulation

In the summary page, click **Finish** and **Build & Run** to build model and open it automatically in the Emulation control view. More details can be found from the previous example, see chapter 4.4.1.7.

5 TECHNICAL SPECIFICATION

This chapter lists briefly the technical specifications of PROPSIM related to the operating environment. For further technical information, please refer to Product Specification.

5.1 Power supply

Input power requirements	
Voltage and frequency	100 to 240 +/-10% VAC, 50/60 Hz, nominal
Power rating	1500 W MAX (15 A MAX)
Power consumption, typical	
8TRX+8TX channels (1 CU BRF1)	800 W
16TRX+16TX channels (2 CU BRF1)	1200 W
8TRX channels (1CU BRF2)	800 W
8TRX channels (2 CU BRF2)	1200 W
8TRX+8TX channels (1 CU BRF1) and 8TRX channels (1CU BRF2)	1200W

5.2 Mechanical specifications

Dimension and weight	
Dimensions (nominal) (H x W x D)	293mm x 435mm x 600mm (Manufacturing tolerances are considered separately) Fits into 19-inch rack
Weight, Fully loaded configuration Net weight Shipping weight	 41 kg 58.5 kg

5.3 Environmental and climatic requirements (operating)

Operating test environment specifications	
Operating environment	PROPSIM has been designed for use in Overvoltage Category II and Pollution Degree 2 environment. The operating should be limited to stationary use at weather-protected and dust-free locations. PROPSIM has been categorized to Noisy Attended Equipment Room environment device.
Operating temperature	+15 to +30 °C

	Rate of change of temperature <0.5 °C/min After transportation, the temperature of the device must be stabilized at least 6 hours before switching the power on to prevent condensation.
Humidity	5% to 80% non-condensing relative humidity
Altitude	Up to 2000 m

Note that during the use free, unrestricted air flow must be allowed around the device to ensure proper cooling (see Figure 5 Air inflow and outflow on sides and back of the emulator)

5.4 Transportation and storage requirements

Transportation requirements	
Transportation	ETS 300-019-1-2, class 2.2 careful transportation
Air temperature	-25 ... +70°C, unventilated enclosures -25 ... +30°C, when direct transfer of the product between the two given temperatures and condensation is presumed.
Humidity	up to 95%, not combined with rapid temperature changes at +40°C up to 95%, combined with rapid temperature changes at high relative humidity
Storage temperature	-25 ... +70°C, unventilated enclosures

After transportation, the temperature of the device must be stabilized at least 6 hours before switching the power on to prevent condensation.

5.5 Electrical safety regulations

Electromagnetic compatibility (EMC)	
IEC 61326-1:2012	Electrical equipment for measurement, control and laboratory use – EMC requirements – Part 1 General Requirements
EN 55011:2009 + A1:2010, CISPR 11:2009 + A1:2010	Conducted Emission
EN 55011:2009+A1:2010 CISPR 11:2009 + A1:2010	Radiated Emission
EN 61000-4-2:2009 IEC 61000-4-2:2008	Electrostatic Discharge Immunity
EN 61000-4-3:2006 + A1:2008 + A2:2010 IEC 61000-4-3:2006 + A1:2007 + A2:2010	Radiated Field Immunity

EN 61000-4-4:2004 IEC 61000-4-4:2004	Fast Transient Immunity
EN 61000-4-5:1995 +A1:2001 IEC 61000-4-5:2005	Surge Immunity
EN 61000-4-6:2009 IEC 61000-4-6:2008	Conducted Immunity
EN 61000-4-8:2010 IEC 61000-4-8:2009	Power Frequency Magnetic Field Immunity
EN 61000-4-11:2004 IEC 61000-4-11:2004	Voltage Dips, Variations and Short Interruptions

Safety	
EN 61010-1:2010 (3rd edition) IEC 61010-1:2010	Basic Safety Standard
CAN/CSA-C22.2 No.61010-1 / UL 61010-1:2012	Deviations to 3rd edition for Canada / USA
Compatible standards and regulations	
WEEE according to EU regulations in case of product disposal, 2011/65/EU	

6 MAINTENANCE

PROPSIM is designed to require minimum maintenance. The main points of maintenance are discussed below.

6.1 Calibration

PROPSIM factory calibration period is 12 months. Calibration expiration date for PROPSIM is visible in the **Device Information** dialog (select **Configuration** > **Device Information** in the Navigation bar).

6.2 Connectors and cables

To maintain the high performance of the emulator, proper care must be taken when connecting cables to the emulator. Maximum torque is 0.8 Nm on the SMA connectors.

It is highly recommended to use SMA torque wrench supplied with the PROPSIM always when connecting RF cables to PROPSIM.

The connectors shouldn't be touched by hand, to prevent dust and dirt going into the connector. Also, possible damages caused by static discharges are avoided.

6.3 Cleaning

The instrument front and rear panels should be cleaned using an anti-static soft cloth. PROPSIM should be kept in a dust-free environment. To prevent electrical shock, disconnect PROPSIM from mains before cleaning. Do not attempt to clean internally.

6.4 Over current protector

The main power switch of the emulator has an over current protector. The emulator has no fuse that the user can change. If over current occurs, the emulator turns itself off and the switch returns to the OFF position.

If the over current protection turns the emulator off repeatedly, it is an indication of some fault in the emulator. The device should be sent to a facility designated by Keysight for appropriate service.

Note: The power switch must not be forced to the ON position if the over current protection shuts down the emulator. It will not help keep the emulator operational.

6.5 Shipment for Service

Contact Keysight Technologies for instructions on where to ship the PROPSIM for service. Ship the PROPSIM using the original packaging materials. Shipping the PROPSIM in anything other than the original packaging may result in non-warranted damage.